Sampling and Reconstruction

2.1

A continuous-time signal $x_c(t) = \cos(2\pi f_0 t)$, where $f_0 = 1$ kHz, is to be sampled and reconstructed.

(a) What is the minimum sampling frequency f_s required to sample $x_c(t)$ without aliasing?

(b) If the signal is sampled at $f_s = 1.5$ kHz, determine the frequency of the aliased component that appears in the reconstructed signal.

(c) Sketch the spectrum of the sampled signal when sampled at $f_s = 1.5$ kHz.

Fourier Transform Properties

2.2

Given the signal $x(t) = e^{-at}u(t)$, where $u(t)$ is the unit step function and $a > 0$:

(a) Find the Fourier Transform $X(f)$ of $x(t)$.

(b) Using the time-shifting property, find the Fourier Transform $Y(f)$ of $y(t) = x(t - t_0)$, where $t_0 > 0$.

(c) Explain how the magnitude and phase of $Y(f)$ relate to those of $X(f)$.

Convolution

2.3

Given two signals:

- $x(t) = \text{rect}\left(\frac{t}{T}\right)$, a rectangular pulse of width T,
- $h(t) = \text{sinc}\left(\frac{t}{T}\right)$, where $\text{sinc}(t) = \frac{\sin(\pi t)}{\pi t}$.

(a) Compute the convolution $y(t) = x(t) * h(t)$.

(b) Explain the result using Fourier Transform properties and convolution.

Aliasing in Sampling

2.4

An analog signal $x_c(t)$ has a frequency spectrum non-zero only between -3 kHz and 3 kHz.

(a) What is the minimum sampling frequency f_s required to sample $x_c(t)$ without aliasing?

(b) If the signal is sampled at $f_s = 4$ kHz, describe and sketch the resulting spectrum $X_s(f)$ of the sampled signal.

(c) Propose a method to prevent aliasing when sampling at $f_s = 4$ kHz.

Image Sampling and Spatial Frequency

2.5:

An image is represented as a continuous function $I(x, y)$, where x and y are spatial coordinates.

(a) Explain how sampling in the spatial domain affects the frequency content of the image.

(b) If the image contains high-frequency components up to spatial frequency $f_{\max}$, what is the minimum sampling rate (samples per unit length) required in x and y directions to prevent aliasing?

(c) Describe what happens if the image is undersampled.

Quantization Levels and Bits

2.6

An analog signal is quantized using a uniform quantizer with N bits per sample.

(a) If the quantizer uses $N = 8$ bits, how many quantization levels M are available?

(b) If the signal amplitude ranges from -1 V to $+1$ V, what is the quantization interval ΔV?

(c) If the number of bits is increased to $N = 10$, how does this affect the quantization interval ΔV and the Signal to Quantization Noise Ratio (SQNR)?

Signal to Quantization Noise Ratio (SQNR)

2.7

A signal with a maximum amplitude $V_{\max} = 5$ V is uniformly quantized using $N = 12$ bits.

(a) Calculate the quantization interval ΔV.

(b) Assuming the quantization error is uniformly distributed between $-\Delta V/2$ and $+\Delta V/2$, calculate the mean squared quantization error σ_{qe}^2.

(c) Determine the SQNR in decibels.

Non-Uniform Quantization and Weber's Law

2.8

Consider a sensory signal where the perception p relates to the stimulus S according to Weber's Law:

$$p = k \cdot \ln\left(\frac{S}{S_0}\right)$$

where S_0 is the threshold stimulus, and k is a constant.

(a) Explain why non-uniform quantization intervals are more suitable for such signals compared to uniform quantization.

(b) If an audio signal follows this logarithmic perception model, how would you design a quantizer to optimize the perceived audio quality?

Distortion Measures in Image Compression

2.9

An 8-bit grayscale image is compressed using lossy compression, resulting in a reconstructed image. The original image has pixel values $I_{\text{original}}(i, j)$, and the reconstructed image has pixel values $I_{\text{reconstructed}}(i, j)$.

(a) Define the Mean Squared Error (MSE) between the original and reconstructed images.

(b) If the peak pixel value is 255, express the Peak Signal to Noise Ratio (PSNR) in terms of MSE.

(c) Calculate the PSNR if the MSE is found to be 100.